Practice Question

Perform **Intensity Level Slicing** with range limits $r_1 = 3$ and $r_2 = 5$ for the given 4×4 input image matrix r:

$$\text{Input Image } (r) = \begin{bmatrix} 4 & 3 & 5 & 2 \\ 3 & 6 & 4 & 6 \\ 2 & 2 & 6 & 5 \\ 7 & 6 & 4 & 1 \end{bmatrix}$$

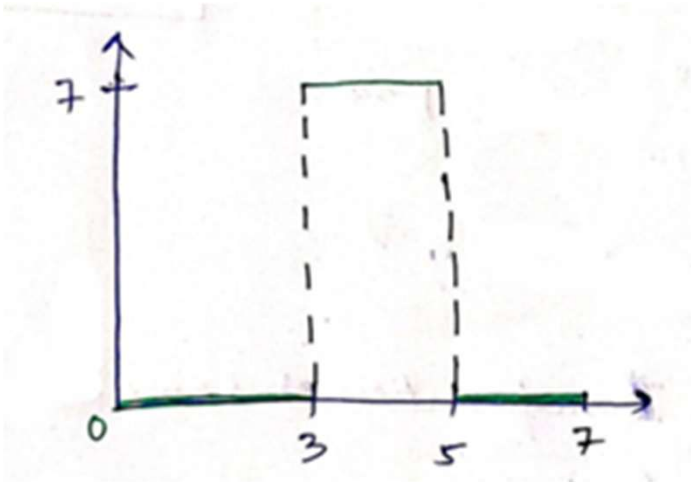
Compute the output image matrix S and sketch the transformation functions for both cases:

1. **Without Background Retention**
2. **With Background Retention**

Case 1: Without Background Retention

- **Transformation Function:**

$$S = \begin{cases} L - 1 = 7, & \text{for } 3 \le r \le 5 \\ 0, & \text{otherwise} \end{cases}$$



- **Logic:**
  - If the pixel value $r \in \{3,4,5\}$, set output value $S = 7$.
  - If $r \in \{0,1,2,6,7\}$, set output value $S = 0$.

- **Resulting Output Matrix (S):**

$$r = \begin{bmatrix} 4 & 3 & 5 & 2 \\ 3 & 6 & 4 & 6 \\ 2 & 2 & 6 & 5 \\ 7 & 6 & 4 & 1 \end{bmatrix} \longrightarrow S = \begin{bmatrix} 7 & 7 & 7 & 0 \\ 7 & 0 & 7 & 0 \\ 0 & 0 & 0 & 7 \\ 0 & 0 & 7 & 0 \end{bmatrix}$$

Case 2: With Background Retention

Transformation Function:

$$S = \begin{cases} L - 1 = 7, & \text{for } 3 \leq r \leq 5 \\ r, & \text{otherwise} \end{cases}$$

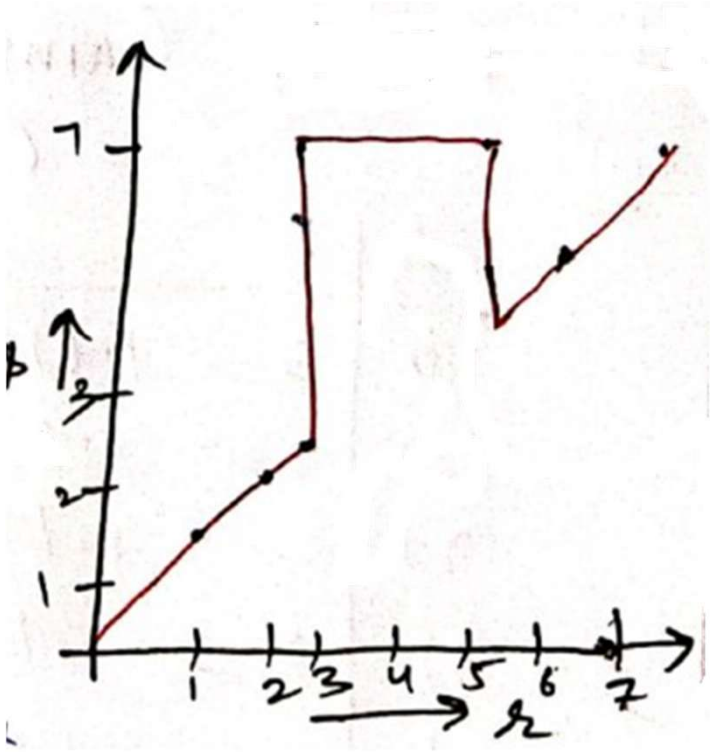
Logic:

If the pixel value $r \in \{3,4,5\}$, map it to $S = 7$.

For all other values, retain the original pixel intensity ($S = r$).

Resulting Output Matrix (S):

$$r = \begin{bmatrix} 4 & 3 & 5 & 2 \\ 3 & 6 & 4 & 6 \\ 2 & 2 & 6 & 5 \\ 7 & 6 & 4 & 1 \end{bmatrix} \longrightarrow S = \begin{bmatrix} 7 & 7 & 7 & 2 \\ 7 & 6 & 7 & 6 \\ 2 & 2 & 6 & 7 \\ 7 & 6 & 7 & 1 \end{bmatrix}$$



Practice Question

A grayscale image contains the following intensity values:

[30, 50, 70, 90, 110]

Normalize the intensity range from **30–110** to **0–255**.

Find the normalized values.

$$s = s_{min} + \frac{s_{max} - s_{min}}{r_{max} - r_{min}}(r - r_{min})$$

Here,

$$r_{\min} = 30, \quad r_{\max} = 110$$

and

$$s_{\min} = 0, \quad s_{\max} = 255$$

The equation becomes:

$$s = \frac{255}{110 - 30}(r - 30)$$

$$s = \frac{255}{80}(r - 30)$$

r	Calculation	Normalized s
30	255/80*(30−30)	**0**
50	255/80*(20)	**63.75 ≈ 64**
70	255/80*(40)	**127.5 ≈ 128**
90	255/80*(60)	**191.25 ≈ 191**
110	255/80*(80)	**255**

Therefore:
[30,50,70,90,110]→[0,64,128,191,255]

Practice Question

Consider the following grayscale image:

$$f(x,y) = \begin{bmatrix} 4 & 3 & 5 & 2 \\ 3 & 6 & 4 & 6 \\ 2 & 2 & 6 & 5 \\ 7 & 6 & 4 & 1 \end{bmatrix}$$

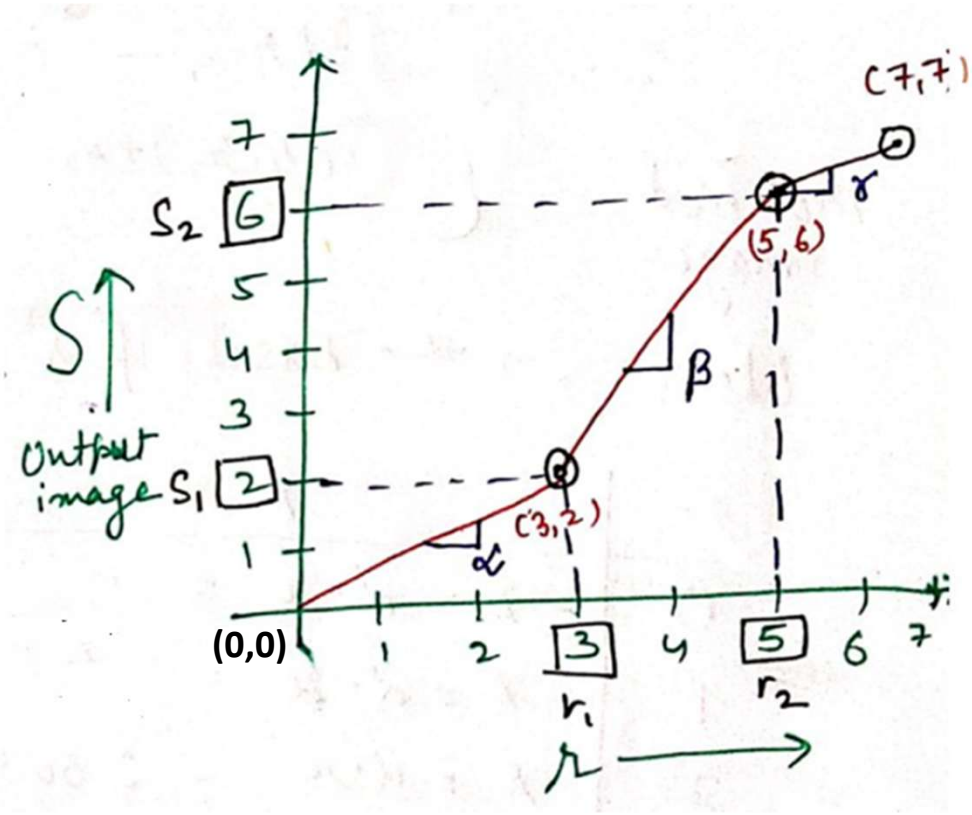
Perform contrast stretching using a piecewise linear transformation with parameters $r_1 = 3$, $r_2 = 5$, $s_1 = 2$, and $s_2 = 6$. Derive the transformation function, compute the transformed value of each pixel, and obtain the final enhanced image matrix.

Parameters:

- $r_1 = 3,\ r_2 = 5$
- $s_1 = 2,\ s_2 = 6$
- 3-bit image $\rightarrow$ $L = 8$, intensity range 0 to 7

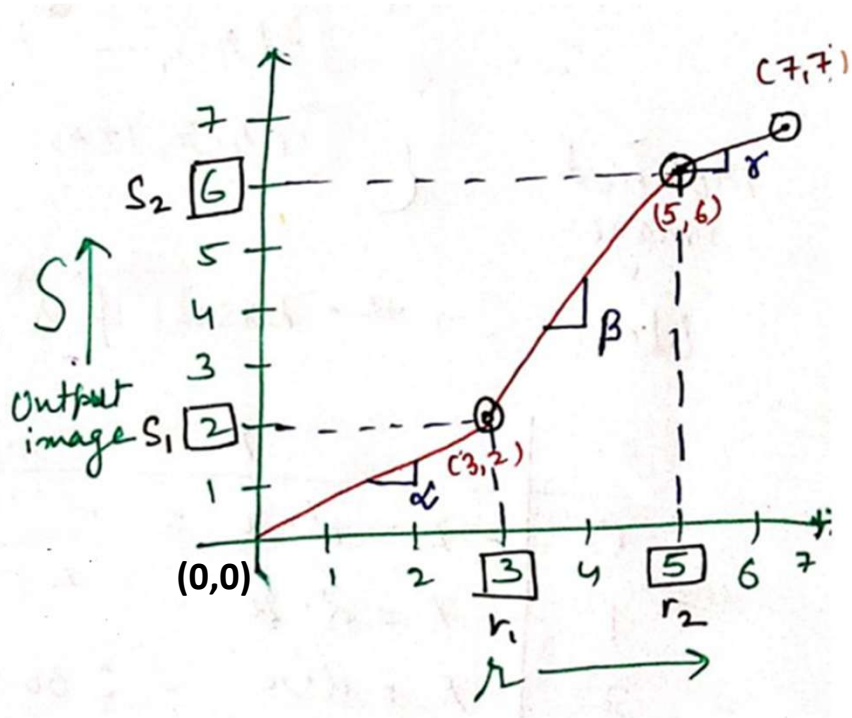
Step 1: Compute Slopes

- $\boldsymbol{\alpha = \frac{s_1 - 0}{r_1 - 0} = \frac{2-0}{3-0} \approx 0.66}$
- $\boldsymbol{\beta = \frac{s_2 - s_1}{r_2 - r_1} = \frac{6-2}{5-3} = \frac{4}{2} = 2}$
- $\boldsymbol{\gamma = \frac{(L-1) - s_2}{(L-1) - r_2} = \frac{7-6}{7-5} = \frac{1}{2} = 0.5}$



Step 2: Piecewise Transformation Function

$$s = \begin{cases} \alpha r, & 0 \le r < 3 \\ \beta(r - r_1) + s_1, & 3 \le r < 5 \\ \gamma(r - r_2) + s_2, & 5 \le r \le 7 \end{cases}$$



Step 3: Compute Mapping (r → s)

r	Calculation	s(approx.)
0	0.66×0	0
1	0.66×1	$0.66 \approx 1$
2	0.66×2	$1.32 \approx 1$
3	$2(3-3)+2$	2
4	$2(4-3)+2$	4
5	$0.5(5-5)+6$	6
6	$0.5(6-5)+6$	$6.5 \approx 7$
7	$0.5(7-5)+6$	7

Piecewise Transformation Function

$$s = \begin{cases} 0.66r, & 0 \le r < 3 \\ 2(r-3)+2, & 3 \le r < 5 \\ 0.5(r-5)+6, & 5 \le r \le 7 \end{cases}$$

Values of

$r = 0,1,2$

$r = 3,4$

$r = 5,6,7$

$$f(x,y) = \begin{bmatrix} 4 & 3 & 5 & 2 \\ 3 & 6 & 4 & 6 \\ 2 & 2 & 6 & 5 \\ 7 & 6 & 4 & 1 \end{bmatrix} \xrightarrow[\text{Stretching}]{\text{Contrast}} g(x,y) = \begin{bmatrix} 4 & 2 & 6 & 1 \\ 2 & 7 & 4 & 7 \\ 1 & 1 & 7 & 6 \\ 7 & 7 & 4 & 1 \end{bmatrix}$$

Practice Question

Given an input image I with dynamic range L:

$$I = \begin{bmatrix} 110 & 120 & 90 \\ 91 & 94 & 98 \\ 90 & 91 & 99 \end{bmatrix}$$

Question 1: Apply Image Inversion (Negative Transformation)

Question 2: Logarithmic Transformation for the following cases:

1. **Case A:** $c = 1$
2. **Case B:** $c = \frac{L}{\log_{10}(1+L)}$

Question 3: Apply Power-Law (Gamma) Transformation assuming $c = 1$and $\gamma = 0.2$.

Question 1: Image Inversion (Negative Transformation)

Using the standard negative formula $s = (L-1) - r = 127 - r$:

$$I_{\text{neg}} = \begin{bmatrix} 127-110 & 127-120 & 127-90 \\ 127-91 & 127-94 & 127-98 \\ 127-90 & 127-91 & 127-99 \end{bmatrix} = \begin{bmatrix} \mathbf{17} & \mathbf{7} & \mathbf{37} \\ \mathbf{36} & \mathbf{33} & \mathbf{29} \\ \mathbf{37} & \mathbf{36} & \mathbf{28} \end{bmatrix}$$

Question 2: Logarithmic Transformation

The logarithmic transformation formula is $s = c \cdot \log_{10}(1 + r)$.

Case A: $c = 1$

Calculating $s = \log_{10}(1 + r)$for each value:

$$I_{\log,\,\text{A}} \approx \begin{bmatrix} \mathbf{2.05} & \mathbf{2.08} & \mathbf{1.96} \\ \mathbf{1.96} & \mathbf{1.98} & \mathbf{2.00} \\ \mathbf{1.96} & \mathbf{1.96} & \mathbf{2.00} \end{bmatrix} \text{ and rounded to integers: } \begin{bmatrix} \mathbf{2} & \mathbf{2} & \mathbf{2} \\ \mathbf{2} & \mathbf{2} & \mathbf{2} \\ \mathbf{2} & \mathbf{2} & \mathbf{2} \end{bmatrix}$$

Question 2: Logarithmic Transformation

Case B: $c = \frac{L}{\log_{10}(1+L)}$
With $L = 128$:

$$c = \frac{128}{\log_{10}(1+128)} = \frac{128}{\log_{10}(129)} \approx \frac{128}{2.1106} \approx 60.646$$

Multiplying each value from Case A by $c \approx 60.646$:

$$I_{\log,\,\mathrm{B}} \approx \begin{bmatrix} \mathbf{124.04} & \mathbf{126.31} & \mathbf{118.80} \\ \mathbf{119.09} & \mathbf{119.94} & \mathbf{121.03} \\ \mathbf{118.80} & \mathbf{119.09} & \mathbf{121.29} \end{bmatrix} \quad \text{and rounded to integers:} \quad \begin{bmatrix} \mathbf{124} & \mathbf{126} & \mathbf{119} \\ \mathbf{119} & \mathbf{120} & \mathbf{121} \\ \mathbf{119} & \mathbf{119} & \mathbf{121} \end{bmatrix}$$

Question 3: Power-Law (Gamma) Transformation
The power-law formula $s = c \cdot r^{\gamma}$ with $c = 1$ and $\gamma = 0.2$ remains unchanged, as it does not depend on L:

$$I_{\mathrm{gamma}} \approx \begin{bmatrix} \mathbf{2.56} & \mathbf{2.60} & \mathbf{2.46} \\ \mathbf{2.46} & \mathbf{2.48} & \mathbf{2.50} \\ \mathbf{2.46} & \mathbf{2.46} & \mathbf{2.51} \end{bmatrix} \text{ and rounded to integers: } \begin{bmatrix} \mathbf{3} & \mathbf{3} & \mathbf{2} \\ \mathbf{2} & \mathbf{2} & \mathbf{3} \\ \mathbf{2} & \mathbf{2} & \mathbf{3} \end{bmatrix}$$

Practice Question

Histogram Equalization

- Example: a **64**-by-**64**, 3-bit (**L=8**) image, with the provided histogram.
- Let's define $\mathbf{s_k}$ as:

$$s_k = (L-1)\sum_{j=0}^{k} p_r(r_j), \text{ for } k = 0, 1, \ldots, L-1$$

- We will have:

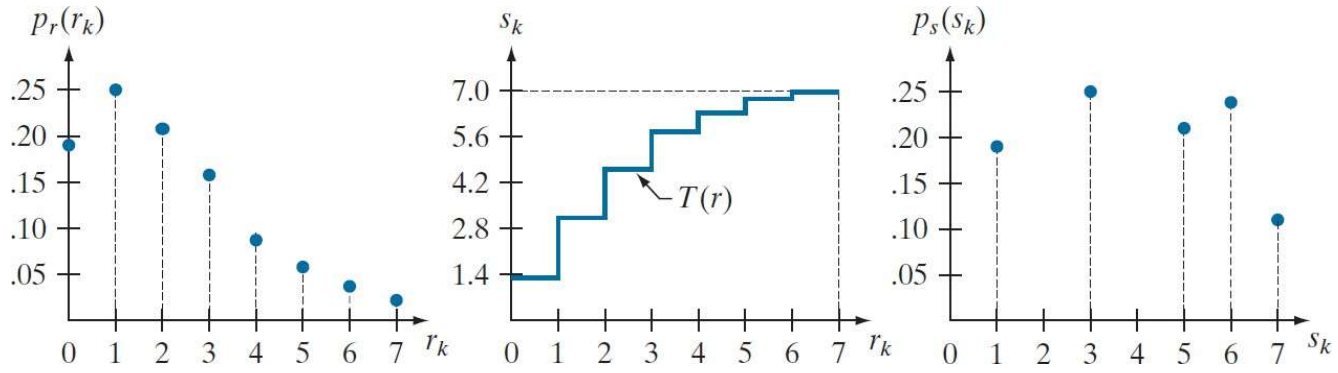
$$s_0 = 7\, p_r(r_0) = 1.33$$
$$s_1 = 7\,(p_r(r_0) + p_r(r_1)) = 3.08$$
$$s_2 = 7\,(p_r(r_0) + p_r(r_1) + p_r(r_2)) = 4.55$$
$$s_3 = \ldots$$

r_k	n_k	$p_r(r_k) = n_k / MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

Histogram Equalization

- Fractional **s** values need to be rounded to the closest integer value in the range **[0,7]**.

$$s_0 = 1.33 \rightarrow 1 \qquad s_4 = 6.23 \rightarrow 6$$
$$s_1 = 3.08 \rightarrow 3 \qquad s_5 = 6.65 \rightarrow 7$$
$$s_2 = 4.55 \rightarrow 5 \qquad s_6 = 6.86 \rightarrow 7$$
$$s_3 = 5.67 \rightarrow 6 \qquad s_7 = 7.00 \rightarrow 7$$



Practice Question

Perform histogram equalisation of the image $\begin{bmatrix} 4 & 4 & 4 & 4 & 4 \\ 3 & 4 & 5 & 4 & 3 \\ 3 & 5 & 5 & 5 & 3 \\ 3 & 4 & 5 & 4 & 3 \\ 4 & 4 & 4 & 4 & 4 \end{bmatrix}$.

Histogram Equalization

Solution The maximum value is found to be 5. We need a minimum of 3 bits to represent the number. There are eight possible gray levels from 0 to 7. The histogram of the input image is given below:

Gray level	0	1	2	3	4	5	6	7
Number of pixels	0	0	0	6	14	5	0	0

Step 1 Compute the running sum of histogram values.

The running sum of histogram values is otherwise known as cumulative frequency distribution.

Gray level	0	1	2	3	4	5	6	7
Number of pixels	0	0	0	6	14	5	0	0
Running sum	0	0	0	6	20	25	25	25

Histogram Equalization

Step 2 Divide the running sum obtained in Step 1 by the total number of pixels. In this case, the total number of pixels is 25.

Gray level	0	1	2	3	4	5	6	7
Number of pixels	0	0	0	6	14	5	0	0
Running sum	0	0	0	6	20	25	25	25
Running Sum / Total number of pixels	0/25	0/25	0/25	6/25	20/25	25/25	25/25	25/25

Step 3 Multiply the result obtained in Step 2 by the maximum gray-level value, which is 7 in this case.

Histogram Equalization

Gray level	0	1	2	3	4	5	6	7
Number of pixels	0	0	0	6	14	5	0	0
Running Sum	0	0	0	6	20	25	25	25
Runningsum / Total number of pixels	0/25	0/25	0/25	6/25	20/25	25/25	25/25	25/25
Multiply the above result by maximum gray level	$\frac{0}{25}*7$	$\frac{0}{25}*7$	$\frac{0}{25}*7$	$\frac{6}{25}*7$	$\frac{20}{25}*7$	$\frac{25}{25}*7$	$\frac{25}{25}*7$	$\frac{25}{25}*7$

Histogram Equalization

The result is then rounded to the closest integer to get the following table:

Gray level	0	1	2	3	4	5	6	7
Number of pixels	0	0	0	6	14	5	0	0
Running Sum	0	0	0	6	20	25	25	25
Running Sum/Total number of pixels	0/25	0/25	0/25	6/25	20/25	25/25	25/25	25/25
Multiply the above result by maximum gray level	0	0	0	2	6	7	7	7

Step 4 Mapping of gray level by a one-to-one correspondence:

Original gray level	Histogram equalised values
0	0
1	0
2	0
3	2
4	6
5	7
6	7
7	7

Histogram Equalization

The original image and the histogram equalised image are shown side by side.

$$\begin{bmatrix} 4 & 4 & 4 & 4 & 4 \\ 3 & 4 & 5 & 4 & 3 \\ 3 & 5 & 5 & 5 & 3 \\ 3 & 4 & 5 & 4 & 3 \\ 4 & 4 & 4 & 4 & 4 \end{bmatrix} \xrightarrow{\text{Histogram Equalisation}} \begin{bmatrix} 6 & 6 & 6 & 6 & 6 \\ 2 & 6 & 7 & 6 & 2 \\ 2 & 7 & 7 & 7 & 2 \\ 2 & 6 & 7 & 6 & 2 \\ 6 & 6 & 6 & 6 & 6 \end{bmatrix}$$

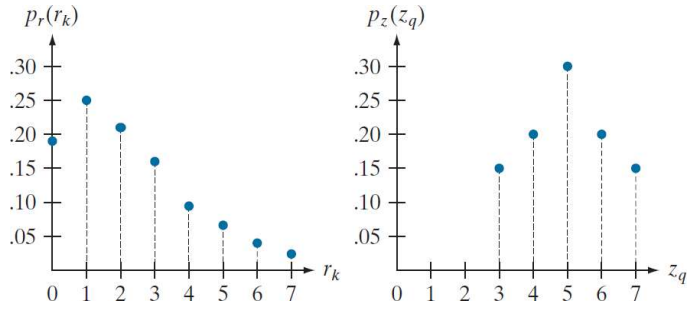
Original image **Histogram equalised image**

Practice Question

Histogram Matching (Specification)

- Example: same **64**-by-**64**, 3-bit (**L=8**) image as the previous example.

r_k	n_k	$p_r(r_k) = n_k/MN$	z_q	Specified $p_z(z_q)$
$r_0 = 0$	790	0.19	$z_0 = 0$	0.00
$r_1 = 1$	1023	0.25	$z_1 = 1$	0.00
$r_2 = 2$	850	0.21	$z_2 = 2$	0.00
$r_3 = 3$	656	0.16	$z_3 = 3$	0.15
$r_4 = 4$	329	0.08	$z_4 = 4$	0.20
$r_5 = 5$	245	0.06	$z_5 = 5$	0.30
$r_6 = 6$	122	0.03	$z_6 = 6$	0.20
$r_7 = 7$	81	0.02	$z_7 = 7$	0.15



Histogram Matching (Specification)

- The first step is to compute the rounded histogram-equalized values, as we did in the previous example:

$$s_0 = 1.33 \rightarrow 1 \quad s_4 = 6.23 \rightarrow 6$$
$$s_1 = 3.08 \rightarrow 3 \quad s_5 = 6.65 \rightarrow 7$$
$$s_2 = 4.55 \rightarrow 5 \quad s_6 = 6.86 \rightarrow 7$$
$$s_3 = 5.67 \rightarrow 6 \quad s_7 = 7.00 \rightarrow 7$$

- Then, we compute the values of $\mathbf{G(z_q)}$ using the values of $\mathbf{p_z(z_q)}$:

$$G(z_0) = 0.00 \quad G(z_2) = 0.00 \quad G(z_4) = 2.45 \quad G(z_6) = 5.95$$
$$G(z_1) = 0.00 \quad G(z_3) = 1.05 \quad G(z_5) = 4.55 \quad G(z_7) = 7.00$$

- These values are then rounded to their nearest integer values:

$$G(z_0) = 0.00 \rightarrow 0 \qquad G(z_4) = 2.45 \rightarrow 2$$
$$G(z_1) = 0.00 \rightarrow 0 \qquad G(z_5) = 4.55 \rightarrow 5$$
$$G(z_2) = 0.00 \rightarrow 0 \qquad G(z_6) = 5.95 \rightarrow 6$$
$$G(z_3) = 1.05 \rightarrow 1 \qquad G(z_7) = 7.00 \rightarrow 7$$

Histogram Matching (Specification)

- The first three values of **G** are equal, so it is not strictly monotonic, therefore condition (**a'**) is violated.
- Given this, we follow step three of the histogram matching approach which states we need to find the smallest value of $\mathbf{z_q}$ so that the value $\mathbf{G(z_q)}$ is the closest to $\mathbf{s_k}$. This is done for every value of $\mathbf{s_k}$ to create the required mapping from **s** to **z**.
- For example, for $\mathbf{s_0=1}$, we have $\mathbf{G(z_3)=1}$ which is a perfect match, which means that every pixel in the histogram-equalized image with value of **1**, is mapped to a pixel with value **3** in the histogram-specified image.

s_k	$\rightarrow$	z_q
1	$\rightarrow$	3
3	$\rightarrow$	4
5	$\rightarrow$	5
6	$\rightarrow$	6
7	$\rightarrow$	7

Histogram Matching (Specification)

r_k	s_k
0	1
1	3
2	5
3	6
4	6
5	7
6	7
7	7



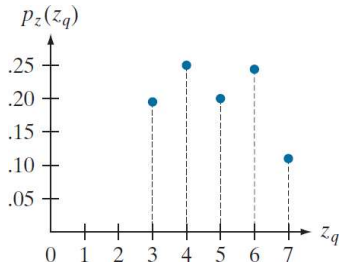
s_k	→	z_q
1	→	3
3	→	4
5	→	5
6	→	6
7	→	7

r_k	O_k
0	3
1	4
2	5
3	6
4	6
5	7
6	7
7	7

Histogram Matching (Specification)

- The final step is to use the computed mappings to map every pixel in the histogram-equalized image into the corresponding pixel in the newly created histogram-specified image.
- The final result does not match the specified histogram exactly, however the general trend was achieved.
- Obtaining the histogram-equalized image is not necessary. We can list the mappings from **r** to **s**, and from **s** to **z**, and then use these mappings to map the original pixels directly into the pixels of the histogram-specified image.

z_q	Specified $p_z(z_q)$	Actual $p_z(z_q)$
$z_0 = 0$	0.00	0.00
$z_1 = 1$	0.00	0.00
$z_2 = 2$	0.00	0.00
$z_3 = 3$	0.15	0.19
$z_4 = 4$	0.20	0.25
$z_5 = 5$	0.30	0.21
$z_6 = 6$	0.20	0.24
$z_7 = 7$	0.15	0.11



Practice Question

Numerical Question: Histogram Matching (Histogram Specification)
Consider the following **5 × 5 grayscale image**:

$$f(x, y) = \begin{bmatrix} 4 & 4 & 4 & 4 & 4 \\ 3 & 4 & 5 & 4 & 3 \\ 3 & 5 & 5 & 5 & 3 \\ 3 & 4 & 5 & 4 & 3 \\ 4 & 4 & 4 & 4 & 4 \end{bmatrix}$$

Desired (Reference) Image
The desired image $g(x, y)$is given as:

$$g(x, y) = \begin{bmatrix} 1 & 2 & 2 & 2 & 1 \\ 2 & 3 & 4 & 3 & 2 \\ 2 & 4 & 6 & 4 & 2 \\ 2 & 3 & 4 & 3 & 2 \\ 1 & 2 & 2 & 2 & 1 \end{bmatrix}$$

Solution:

r_k	n_k	p_r(r_k)
0	0	0
1	0	0
2	0	0
3	6	0.24
4	14	0.56
5	5	0.2
6	0	0
7	0	0

$$s_0 = 0 \rightarrow 0$$

$$s_1 = 0 \rightarrow 0$$

$$s_2 = 0 \rightarrow 0$$

$$s_3 = 1.68 \rightarrow 2$$

$$s_4 = 5.6 \rightarrow 6$$

$$s_5 = 7 \rightarrow 7$$

$$s_6 = 7 \rightarrow 7$$

$$s_7 = 7 \rightarrow 7$$

Solution:

r_k	n_k	p_r(r_k)
0	0	0
1	4	0.16
2	12	0.48
3	4	0.16
4	4	0.16
5	0	0
6	1	0.04
7	0	0

$$G(z_0) = 0 \rightarrow 0$$

$$G(z_1) = 1.12 \rightarrow 1$$

$$G(z_2) = 4.48 \rightarrow 4$$

$$G(z_3) = 5.6 \rightarrow 6$$

$$G(z_4) = 6.72 \rightarrow 7$$

$$G(z_5) = 6.72 \rightarrow 7$$

$$G(z_6) = 7 \rightarrow 7$$

$$G(z_7) = 7 \rightarrow 7$$

Solution:

r_k	s_k	Z_q
3	2	1
4	6	3
5	7	6

$$f(x,y) = \begin{bmatrix} 4 & 4 & 4 & 4 & 4 \\ 3 & 4 & 5 & 4 & 3 \\ 3 & 5 & 5 & 5 & 3 \\ 3 & 4 & 5 & 4 & 3 \\ 4 & 4 & 4 & 4 & 4 \end{bmatrix}$$

Final histogram-matched image

$$\begin{bmatrix} 3 & 3 & 3 & 3 & 3 \\ 1 & 3 & 6 & 3 & 1 \\ 1 & 6 & 6 & 6 & 1 \\ 1 & 3 & 6 & 3 & 1 \\ 3 & 3 & 3 & 3 & 3 \end{bmatrix}$$

Practice Question

Histogram Stretching

Given

Image size

$$8 \times 8 = 64 \; pixels$$

Gray levels

$$r_k = 0,1,2,3,4,5,6,7$$

Gray level (r_k)	0	1	2	3	4	5	6	7
Number of pixels (n_k)	0	0	5	20	20	19	0	0

Total pixels check:

$$5 + 20 + 20 + 19 = 64$$

Histogram Stretching

1. Understand Histogram Stretching

$$s_k = \frac{r_k - r_{\min}}{r_{\max} - r_{\min}} \cdot (s_{\max} - s_{\min}) + s_{\min}$$

Where:

$r_{\min} = 2$(minimum gray level present with $n_k > 0$)

$r_{\max} = 5$(maximum gray level present with $n_k > 0$)

$s_{\min} = 0$(minimum desired output gray level)

$s_{\max} = 7$(maximum desired output gray level)

Histogram Stretching

2. Step-by-Step Calculations for Existing Gray Levels

For $r_k = 2$:

$$s_2 = \frac{2-2}{5-2} \times (7-0) + 0 = \frac{0}{3} \times 7 = \mathbf{0}$$

For $r_k = 3$:

$$s_3 = \frac{3-2}{5-2} \times (7-0) + 0 = \frac{1}{3} \times 7 \approx 2.33 \Longrightarrow \mathbf{2}$$

For $r_k = 4$:

$$s_4 = \frac{4-2}{5-2} \times (7-0) + 0 = \frac{2}{3} \times 7 \approx 4.67 \Longrightarrow \mathbf{5}$$

For $r_k = 5$:

$$s_5 = \frac{5-2}{5-2} \times (7-0) + 0 = \frac{3}{3} \times 7 = \mathbf{7}$$

Histogram Stretching

3. Mapping Table

Original (r_k)	Pixels (n_k)	New (s_k)
2	5	0
3	20	2
4	20	5
5	19	7

Histogram Stretching

4. Construct Stretched Histogram

New gray level (s)	0	1	2	3	4	5	6	7
Pixels	5	0	20	0	0	20	0	19